FALCON 4K

2.3.0-OA Changelog:

- **ONE FILE INSTALL**: automatic installer, install and boot on Falcon

- Added new F4WHN v7.0.0 and UVStudio v2.2.0 compatibility - Up to 64 apps

- Added VOX function in **Menu**

- Added Journal page (short *) in History: 512 permanent entries, recording VFO RX TX events, in VFO, VFO scan, Full Watch, and Multilist Watch. Shows frequency memory names. Shares same memory zone and RF Log format than F4WHN: journal entries can be seen in both firmware and extracted in UVStudio booting in F4WHN. See *# Keys Infos* on radio

- Added History F1/F2/M Long shortcut in VFO: No RX while browsing Frequency History from VFO shortcut. Frequency History scan and all History functions remains.

- **Full Watch mode replaces Triple & Quad Watch**, with 4 priority channels instead of 2. Full Watch mode can now listen up to 6 channels with VFO A and B.
F+1 F+2 F+3 F+4 saves/removes a priority channel directly from the VFO.
RX detection speed has also been improved.

- F+4 to make a channel being monitored is replaced by F+EXIT

- **Added PONSCR** to **Menu**: Choose on which screen/mode the radio start

- **Added PTT2** to the programmable **M Long shortcut**.

- **New yellowish Led color** during scan operation

- New firmware architecture with cleaner code. Flash memory: 91.50%

Small other fixes:
- Fixed Spectre Record Trigger linked to DS value in Sentinel
- Fixed MHz entry validation: 5, 50 and 500 now confirm as 5, 50 and 500 MHz.
- Fixed explicit decimal entry, including four-digit MHz frequencies such as 1000*0.
- Removed parameter-row visual artifacts during frequency entry in Dual VFO
- Added short EXIT backspace and long EXIT cancellation to Channel/Band/List editors.
- Positioned the name cursor on the last existing character when entering the NAME field.
- Fixed navigation inversion in editors and Edit/New/Delete popups.
- Prevented long EXIT release from triggering another action after leaving an editor.
- Unified Range editor frequency parsing and backspace behavior.
- Fixed Range arrows to preserve interval width, respect frequency limits and avoid band-name popups.

Installation

Main slot: flash **Falcon4K_v3.0.0-OA.bin** with UV Studio in DFU mode, then wait for the automatic install to finish after the reboot.

Multiboot slot: boot on F4HWN (or a UV Studio compatible multiboot firmware), delete the old Falcon slot if there is one, install Falcon on a free slot (or the same one), turn off, boot on Falcon and wait for the automatic install to finish.

> *Falcon v3.0.0 is tested with the F4HWN 7.0.0 memory architecture. A future F4HWN update that changes it could disturb multiboot or the UV Studio online apps.*

Falcon Apps included:

- **FGuide**: All Falcon keys & shortcuts Guide in an App
- **FSystemInfos**: Falcon compatible System Informations App

User guide v3.0.0

Key reference + detailed guide

Firmware for the Quansheng **UV-K5 (V3)** and **UV-K1**, derived from Sonic and compatible with **F4HWN v7** and UV Studio.

Notation: Long X = press and hold X. F + X = short press on F, then on X. ◄ / ► = arrow keys. SK1 / SK2 = side keys (F1 / F2). F (#) opens the “Keys Infos” popup on the screens that offer it.

KEY REFERENCE

1. Main VFO keys

Key	Function
0 – 9	Enter a frequency (frequency mode) or a channel number (memory mode). Six digits = XXX.xxx MHz, confirmed automatically; MENU confirms sooner.
*** (during entry)**	Inserts the decimal point (e.g. 1000 * 0 = 1000.0 MHz).
◄ / ►	Frequency mode: tunes by the selected step. Memory mode: browses the channels, grouped by list.
MENU	Opens the main menu. During an entry: confirms the frequency.
EXIT	Deletes the last digit entered. With the backlight off, the first press only wakes it.
Long EXIT	Starts a new frequency entry (clears the current entry).
Long 1	Memory channel → frequency mode, on the same frequency and with the same settings.
Long 2	Switches VFO A ↔ B (DUAL, FULL WATCH and ML WATCH).
Long 3	Switches frequency mode ↔ memory mode.
Long 4	Cycles the step. In memory mode, the new step is saved to the channel.
Long 5	Switches WIDE / NARROW (FM Narrow follows SetNFM).
Long 6	Cycles the TX power (LOW 1–5, MID, HIGH).
Long 7	Turns Falcon mode on / off (SINGLE or DUAL).
Long 8	Squelch 1 → 9, looping. Level 0 and fine adjustment: Sql menu.
Long 9	Backlight ON / OFF, or on demand.
Long 0	Modulation FM → AM → USB.
Long MENU	Action set in the “M Long” menu (default: flashlight; repeat presses within 3 s to change mode).
Long *	On a memory channel: scans its list at normal speed, with the Monitor frequencies. A and B in frequency mode (DUAL): scans the range between A and B.
Long F	Keypad lock (SetLck: keys only, or keys + PTT).

> *A long press on a digit cancels the current entry before acting. In SINGLE, Long 2 has no effect. In the status bar, SW / DW / FW / MW shows the receive mode (Single, Dual, Full Watch, ML Watch) and VX shows that VOX is on.*

2. F + key shortcuts

Combination	Function
F + MENU	In frequency mode: creates a channel from the current settings. On a memory channel: EDIT / NEW / DEL CH. On a band slot (4067–4096): band actions (BND).
F + EXIT	Memory channels only: adds / removes its frequency from the Monitor group (“MONI ADD” / “MONI DEL”, 50 at most). No effect on an unsaved VFO frequency.
F + 0	Enters / leaves the FM radio.
F + 1 to F + 4	Makes the displayed memory channel priority channel PRICH1 to PRICH4 (“CH PRI n”), or removes it if it already is.
F + 5	Apps menu.
F + 6	Opens Sentinel (LIST / BAND / RANGE, switched with key 6).
F + 7 / 8 / 9	Opens Spectrum in LIST / BAND / RANGE mode.
F + ◄ / ►	Adjusts the VFO dBm squelch threshold (SQLSCN). The value is shown for a moment.
F + SK1 / SK2	Increases / decreases the step (saved to the channel in memory mode).
F + *	On a memory channel: fast scan (FastScan) of all eligible channels, excluding the Monitor group and the bands. In frequency mode: fast scan from the VFO frequency, with its modulation, bandwidth and step.

> *Press F briefly: the F icon appears, then press the key you want. Priority channels are used by Full Watch mode (see Detailed guide, section 2).*

3. Programmable side keys and M Long

Menus **F1Shrt, F1Long, F2Shrt, F2Long** (short / long press on SK1 and SK2) and **M Long** (long press on MENU). Available actions:

Action	Function
NONE	No action.
FLASHLIGHT	Flashlight (ON, SOS, BEACON modes set in the LIGHT menu).
POWER	Cycles the TX power.
MONITOR	Opens the squelch while the key is held.
FM RADIO	Enters / leaves the FM radio.
LOCK KEYPAD	Locks / unlocks the keypad.
FREQ / MEM	Switches frequency mode ↔ memory mode.
MODE	Changes the modulation.
PTT TOGGLE	Switches the PTT mode: hold to talk, or press once to transmit and once to stop.
WIDE NARROW	Switches WIDE / NARROW.
ScanLst Up / Dw	Moves to the next / previous memory list.
RXMOD	Cycles SINGLE → DUAL → FULL WATCH → ML WATCH (unavailable modes are skipped).
SCAN / SCAN ALL	Same functions as Long * and F + * in the VFO.
PTT 2	Transmits on the unselected VFO; the main PTT stays on the active VFO.
HISTORY	Opens the History from the VFO, without listening; * switches to the Journal.
VOX	Turns VOX on / off.
VFO A/B	Switches VFO A ↔ B (not offered for M Long).

> *Factory settings: F1 short = ScanLst Up, F1 long = MONITOR, F2 short = ScanLst Dw, F2 long = RXMOD, M Long = FLASHLIGHT.*

4. Spectrum and Sentinel

Key	Function
1	Skips the current reception (temporary, nothing is deleted).
2	Backlight ON / OFF. ON: follows BLTime and lights up on RX and key presses. OFF: stays off until 2 is pressed again or the scan is reopened.
3	Speed Normal ↔ FastScan.
5	Detail screen for the current or last received frequency.
6	Cycles LIST → BAND → RANGE in the current view.
7	Saves the scanner settings.
9	Opens the History.
0	Changes the modulation for the scan session (channels and bands are not modified).
*** / F**	Spectrum: raises / lowers the dBm squelch (dotted line). Sentinel: raises / lowers the relative squelch (values loop).
◄ / ►	Previous / next list or band. In RANGE: shifts the whole range by its own width.
SK1	NORMAL → FREQ LOCK (stays on the last received frequency) → MONITOR (squelch open).
SK2	Blacklists the received frequency and resumes the scan.
MENU	List or band selection; in RANGE, the SCANRANGE editor.
Long MENU	Scanner parameters page (F (#) for help).
PTT	Transmits according to the PTT option: CHANNEL (VFO selected before entering) or LAST RX (last received frequency). PTT 2 has its own option.
EXIT	Back to the VFO.

> *Spectrum applies a dBm threshold on top of relative detection; Sentinel uses relative detection only. Each mode keeps its own settings and its own selections. Remember to press 7 to keep your settings.*

5. List and band selection

Key	Function
◄ / ►	Browses the entries (selected entries are shown first).
4	Adds / removes the highlighted entry from the multiple selection.
MENU	Selects only this entry and starts the scan.
Long MENU	EDIT / NEW / DEL for the list or band.
*****	Select all / deselect all.
EXIT	Starts scanning the multiple selection (empty: “SELECT A LIST” or “SELECT A BAND”).
F (#)	Keys Infos.

6. History

Key	Function
9 (in the scanner)	Opens the History: up to 200 detected frequencies, with received code, duration, reception count and blacklist.
◄ / ►	Selects an entry and listens to it with its own modulation and bandwidth; the scan pauses.
MENU	Menu: CLEAR HISTORY, CLEAR BLACKLIST, CLEAR NON BLACKLIST, SAVE HISTORY, SAVE HIST TO CH (run with MENU or 1).
Long MENU	Creates a prefilled channel (frequency, step, modulation, bandwidth, code); MENU saves it.
*****	Switches to the Journal.
Long *	Starts / stops scanning the History frequencies.
6	Deletes the entry (an already saved channel is not affected).
Long 1	Sends the entry to the VFO in frequency mode, with its modulation, bandwidth and code, and closes the scanner.
SK2	Adds / removes the entry from the blacklist.
SK1	NORMAL → FREQ LOCK → MONITOR.
0	Changes the modulation.
F (#)	Keys Infos.
EXIT	During the History scan: stops it and stays on the page. Otherwise: back to the scanner, or to the VFO.

> *Opened from Spectrum or Sentinel, the History is shown over the scanner, which keeps running: new entries keep arriving as long as none is selected. Opened from the VFO (HISTORY action), it listens to nothing: not in the background, not on the selected entry, and the LED stays off; only the History scan (Long *) listens. SAVE HISTORY keeps the History for later sessions; SAVE HIST TO CH copies every entry into free channels.*

7. Journal

Key	Function
*** (in the History)**	Opens the Journal; * returns to the History.
◄ / ►	Selects an entry.
F then ◄ / ►	Jumps to the newest or oldest entry.
MENU	FILTER: ALL / RX / TX (active filter marked “<====”, MENU confirms and returns to the Journal). CLEAR JOURNAL erases everything after confirmation (SURE?, second press on MENU).
Long MENU	Saves the frequency to a channel (prefilled editor).
6	Deletes the entry.
Long 1	Sends the frequency to the VFO, with the recorded settings.
EXIT	Leaves (back to the scanner or the VFO).

> *The Journal keeps 512 entries in permanent memory, in the F4HWN RF Log format. It records RX of at least 1 s and TX of the radio in VFO and memory mode, including during scans started from the VFO, in Full Watch and in ML Watch. Spectrum and Sentinel RX and TX are not recorded there (they stay in the History). Each row shows the channel number and name (or the frequency), the S level or “TX” and the power, then the duration. A VFO row whose frequency exists in memory shows the first channel found. A dotted line marks each power-up. No scan, no listening and no LED on this page.*

8. Falcon mode

Action	How to
Turn on	Long 7 in SINGLE or DUAL: 3 green flashes. The backlight and flashlight turn off.
Turn off	Short EXIT or Long 7: normal controls come back.
Switch A / B	In DUAL, Long 2: 1 red flash = VFO A, 2 slower red flashes = VFO B.
Active keys	PTT and Long 2 (in DUAL) only; all other keys are blocked.
Indications	Reception: brief green flash. Transmission: red LED, limited to 1 s.

9. FM radio

Key	Function
F + 0 / EXIT	Enters the FM radio / returns to the VFO.
◄ / ►	Auto mode: next station. Manual mode: 0.1 MHz steps, wrapping at the band limits.
*****	Switches auto seek ↔ manual tuning.
1 – 9	Recalls a favorite (no effect if empty).
Long 1 – 9	Saves the station to that favorite (replaces the previous one).
MENU	Shows / hides the station name.
0	Turns background VFO reception on / off; the FM radio keeps playing.
PTT	Transmits on the radio VFO (never on the FM station), then FM listening resumes.

> *While listening to FM, actions that would change the VFO radio settings are blocked; the flashlight and other compatible actions remain available.*

10. Editors (channel, list, band)

Key	Function
◄ / ►	Previous / next field.
1 / 3	Previous / next value. On NAME: moves the cursor.
4 / 6 (NAME)	Scrolls through letters (hold to repeat).
7 / 9 (NAME)	Scrolls through space, digits and symbols.
Digits (frequency)	Enters the frequency or offset, up to 5 decimal places.
*** (frequency)**	Inserts the decimal point. Enter all 5 decimals to avoid rounding to the step.
MENU	Checks, saves and exits. On error, the editor stays open for correction.
EXIT	Deletes the pending digit; otherwise exits without saving.
Long EXIT	Exits without saving.
F (#)	Keys Infos (not in SCANRANGE).
PTT	Exits without saving and transmits.
Long 4 / Long MENU	In a list's channel list (EDIT LST): puts the channel in Monitor / opens it to manage it.

DETAILED GUIDE

1. Single and dual VFO

Each VFO works in frequency mode (free frequency) or memory mode. **Long 3** switches between the two, keeping the remembered state; **Long 1** copies the displayed channel exactly into frequency mode, with its settings. In DUAL, FULL WATCH and ML WATCH, **Long 2** changes the active VFO (A or B).

Direct six-digit entry works in both layouts. 5, 50 and 500 are confirmed as 5, 50 and 500 MHz; * inserts the decimal point for four-digit MHz frequencies (1000 * 0). In memory mode, the arrows browse the channels grouped by list, without renumbering the memory.

2. Full Watch and Multilist Watch

Set **RxMode = FULL WATCH**: the radio watches A, B and up to four priority channels (PRICH1 to PRICH4), six channels at most. **ML WATCH** scans the memory lists of A and B in the background.

Item	Function
PRICH1 to PRICH4	Ordinary memory channels (not bands) or NONE. Set in the menu or with F + 1 to F + 4 from the VFO.
Display	In Full Watch, priority channels appear as numbered bubbles in the middle of the screen. ML WATCH shows no bubbles.
Reception	A watched channel that receives moves to the active VFO; the one it replaces stays in the watch cycle.
Long 2	Changes the selected VFO A / B, without clearing the priorities.
Cycle	Channels are checked one after another, not at the same time: there is only one receiver.
Restrictions	Falcon mode is only available in SINGLE and DUAL.

Multilist Watch rules

- RXMOD cycles Single → Dual → Full Watch → ML Watch → Single; unavailable modes are skipped.
- With two different lists: list A → Monitor → PRICH1 → … → PRICH4 → list B → Monitor → PRICH1 → … → PRICH4, then again. Each list keeps its position.
- If A and B are in the same list, it is scanned only once (through A).
- A memory without a list selects the unlisted channels. Band slots and internal Monitor copies are excluded.
- A VFO in frequency mode contributes no list; the other one can still be scanned. With no list at all, the radio returns to Dual.
- Valid priorities that are not in the scanned lists are checked between the lists; a duplicate is checked only once.
- At rest, A and B show their original frequency. A reception shows the received channel on its VFO, then the display returns when the reception ends.
- PTT always uses the selected VFO and its transmit settings, never the temporarily received channel.
- Changing VFO does not interrupt reception on the other one; operating the receiving VFO, or EXIT, can interrupt it.
- SQLSCN sets the scan opening threshold; SQLHYS adds a 0 to 8 dBm holding margin (opens at the threshold, closes below threshold − SQLHYS).
- Manual scan, Spectrum, Sentinel, FM radio, menus and apps take over; ML Watch resumes when you return to the VFO screen.
- Falcon mode is turned off when entering ML Watch. The scan runs continuously in FastScan, without the Dual Watch power-saving cycle.

3. Falcon mode

Discreet, battery-saving mode for simple RX / TX exchanges: backlight off, almost all keys blocked, LED indications only. It uses the selected SINGLE or DUAL reception.

4. Spectrum and Sentinel

Mode / view	Description
LIST	Scans the channels of one or more lists. Each channel brings its modulation and step; the listening bandwidth comes from ListenBW.
BAND	Scans one or more saved bands: limits, modulation and step.
RANGE	Scans a temporary range; MENU opens SCANRANGE (FSTART, FSTOP, STEP, MOD, BANDW).
Spectrum	Full-width activity graph, with a footer (selected lists or bands, or the range limits).
Sentinel	Compact display: active frequency, progress or S-meter, last reception, counter. The timer counts the time since the last reception.
Normal / FastScan	Key 3 changes the speed. FastScan measures the signal more briefly; actual speed depends on the scan and the signals.

Scans started from the VFO

Long * scans the channel's list at normal speed; F + * scans all eligible channels in FastScan; in frequency mode, F + * starts from the VFO frequency with its settings. The VFO display is kept: 1 skips the reception, SK2 blacklists the frequency, EXIT or another key stops the scan. With A and B in frequency mode, Long * scans the range between them; the blacklist of this scan is temporary and separate from the History blacklist.

5. History and blacklists

The History keeps up to 200 scanner detections: frequency, received code, blacklist, duration and reception count. VFO receptions in SINGLE, DUAL or FULL WATCH are not added to it: they go to the Journal. SK2 toggles the blacklist, Long MENU creates a prefilled channel, Long 1 sends the entry to the VFO.

6. RX / TX Journal

The Journal is a permanent log of the radio's activity (see the Key reference, section 7). It shares the F4HWN RF Log format: its entries remain readable under F4HWN firmware. The ALL / RX / TX filter in the menu limits the display without erasing anything.

7. VOX

VOX menu: OFF or levels 1 to 10. VOX works in FM and shows "VX" in the status bar; a side key can toggle it with the VOX action. Sensitivity also depends on the microphone gain (Mic menu): if key clicks trigger short transmissions, lower Mic or choose another VOX level.

8. Channel, list and band editors

Create or edit a channel

From a VFO in frequency mode, F + MENU creates a channel from the current settings. From a memory, F + MENU offers EDIT CH, NEW CH and DEL CH: choose with the arrows and confirm with MENU (DEL asks for a second confirmation). A new channel assigned to a list is placed after that list's channels when space is left.

Field	Function
LIST and NAME	List membership (or BAND / MONITOR destination where offered) and channel name (7 characters; lists: 6, bands: 8).
STEP and FREQ	Step and RX frequency. An incomplete entry may be rounded to the step; 5 decimals keep the exact value.
MOD and BAND	FM, AM or USB, then WIDE / NARROW bandwidth.
POWER	LOW 1–5, MID or HIGH (actual output depends on calibration and frequency).
OFFSET	TX offset and direction OFF, + or −.
CTCSS RX / TX	OFF or a tone, separately for receive and transmit; replaces a DCS code in the same direction.
DCS RX / TX	OFF or a normal / inverted DCS code, separately for RX and TX; replaces a CTCSS tone in the same direction.

Create or edit a list

In Spectrum or Sentinel LIST mode, MENU opens the selection; Long MENU offers EDIT LST, NEW LST and DEL LST. NEW asks for a free number from 1 to 25 and a name of up to 6 characters. Changing a list's number moves its channels to it without changing their frequencies. DEL LST deletes the list and its channels, after confirmation. In EDIT LST, Long 4 puts a channel in Monitor or removes it, Long MENU opens it to manage it.

Create or edit a band

In Spectrum or Sentinel BAND mode, Long MENU in the selection offers EDIT BND, NEW BND and DEL BND. The same actions open with F + MENU on memories 4067 to 4096, which hold the 30 band definitions.

Field	Function
Slot and NAME	Band slot and optional name (8 characters). A new band needs a free slot.
STEP	Sweep step.
FSTART / FSTOP	Valid RX limits, FSTOP strictly greater than FSTART.
MOD / BANDW	Modulation and bandwidth used to scan the band.
Save / delete	MENU saves. DEL BND asks for confirmation and does not erase the channels inside the band.

Temporary range (SCANRANGE)

In RANGE, MENU opens SCANRANGE: FSTART, FSTOP, STEP, MOD and BANDW. Arrows change the field, 1 / 3 change the values, MENU applies the range. EXIT first deletes the pending digit, then cancels.

Monitor frequencies and priority channels

On a memory channel, F + EXIT makes a Monitor copy in a free slot, keeping the original channel's list; a second F + EXIT removes it. The group holds 50 frequencies at most ("MONI FULL"), depending on free space; band slots cannot be added. Turn on MONITOR CH in the scanner parameters to include it in scans. F + 1 to F + 4 assign the displayed channel to PRICH1 to PRICH4 the same way.

9. FM radio

Enter with F + 0 or a side key set to FM RADIO. Nine favorites, separate from the radio channels, auto seek or 0.1 MHz steps, station name and reception measurements. VFO reception can stay active in the background (key 0).

10. Falcon-specific menu entries

Menu	Function
RxMode	SINGLE, DUAL, FULL WATCH or ML WATCH.
PRICH1 to PRICH4	Priority channels for Full Watch and ML Watch.
F1Shrt … M Long	Side-key and MENU long-press actions (Key reference, section 3).
POWER	Power LOW 1–5, MID or HIGH.
Sql / SQLSCN / SQLHYS	Squelch in levels (fixed VFO); dBm opening threshold for the VFO and its scans; 0–8 dBm holding margin.
VOX	OFF or levels 1 to 10.
Roger / AUROG	End-of-transmission beep (OFF, STARWARS, R2D2, FALCON, ROGGER, OURO); automatic Roger transmission every 10 s, 30 s, 2, 5, 10 or 30 min.
Tone	OFF or 1750 Hz: SK2 sends the tone while transmitting.
LIGHT	Flashlight mode: ON, SOS, BEACON, OFF.
SetCtr / SetVol / SetMet	Screen contrast; receive volume; S-meter style.
BatTyp	Battery model and capacity (UV-K1 / UV-K5).
ListenBW FM / AM	In the scanner parameters: FM listening bandwidth (25k / 12.5k / 6.25k) and AM (Wide / Narrow).

Appendix: differences from Sonic

Topic	In Falcon
Ultra Watch	Becomes Sentinel, with LIST / BAND / RANGE and a compact display.
Scan access	F + 7 / 8 / 9: Spectrum LIST / BAND / RANGE; F + 6: Sentinel. The SPECTRUM / ULTRA WATCH / SMOOTH SPECTRUM cycle and the key-8 view switch are removed.
History	Compact rows, Keys Infos; Long * scans, MENU opens the actions, Long MENU creates a channel, 6 deletes, SK2 handles the blacklist. Close Call and the old shortcuts are removed.
Lists and bands	25 lists (6-character names) and 30 bands (8 characters) in memories 4067 to 4096.
VFO entry	Direct six-digit entry in single and dual VFO; Long 1 copies the memory, Long 3 switches frequency / memory.
Navigation	Memories grouped by list, without renumbering the memory.
RF filters	F4HWN FM and AM Narrow profiles for the UV-K1; scanner's own AM listening filter.
Display	CTCSS / DCS codes shown instead of AM / FM when useful; F4HWN-style screen refresh.
Startup	The radio restarts on the VFO; the splash screen shows the battery and the CPU temperature.
Additions	On-radio editors, Full Watch, ML Watch, Falcon mode, Monitor group, ListenBW AM, SetVol, AUROG, POWER, SetMet, LIGHT and Tone menus.

Guide written for Falcon 4K v3.0.0 from the Falcon USER GUIDE and the firmware code. Use it as a quick reference: always check the behavior on your radio.